PlotStudio

Framingham 10-year CHD risk factors

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QUESTION

What are the strongest risk factors for developing coronary heart disease within ten years in the framingham dataset?

Data Preparation — Missing Values Handled

The CHD modeling cohort is now ready, and the main crude risk signals are already visible. The cleaned cohort table `df_framingham_model` retains `4,119` participants and `613` 10-year CHD events after dropping only `121` rows (`2.9%`) for low-missingness core predictors, while preserving the full glucose signal through median imputation plus a *glucose missingness* indicator.

Observations

Sample retention was strong. We started with `4,240` adults and `644` CHD events, and ended with `4,119` adults and `613` events, so the analytic sample remains close to the original cohort. Clinically extreme values were retained rather than removed, which is appropriate here because very high blood pressure, cholesterol, BMI, and glucose may reflect the highest-risk patients rather than bad data.

The clearest crude differences were age and blood pressure. In the CHD group, mean *age* was `54.1` vs `48.7` years in the non-CHD group, and mean *systolic BP* was `143.5` vs `130.3` mmHg. *Prevalent hypertension* was also much more common among CHD cases (`50.1%` vs `27.5%`), and *male sex* was more common (`54.8%` vs `41.5%`). Cholesterol and glucose were also higher in the CHD group, while smoking status alone showed only a modest crude gap.

The strongest unadjusted odds ratios were concentrated in older age, vascular/metabolic disease, and BP burden. Crude logistic models showed especially large associations for age `65+` vs `<45` (OR `3.53`), *prevalentStroke* (OR `3.30` , but sparse), *diabetes* (OR `3.15`), *BPMeds* (OR `2.87`), and *prevalentHyp* (OR `2.64`). Among continuous measures, odds rose by about `2.13x` per 10-year increase in age, `1.27x` per 10 mmHg increase in systolic BP, and `1.11x` per 10-unit increase in glucose. Smoking intensity looked more informative than smoker/non-smoker status: *cigsPerDay* per 10 cigarettes had OR `1.15` , while *currentSmoker* alone was weaker and not clearly distinct at this crude stage.

The visual pattern is clinically intuitive and supports the next modeling step. The incidence chart

Figure 1

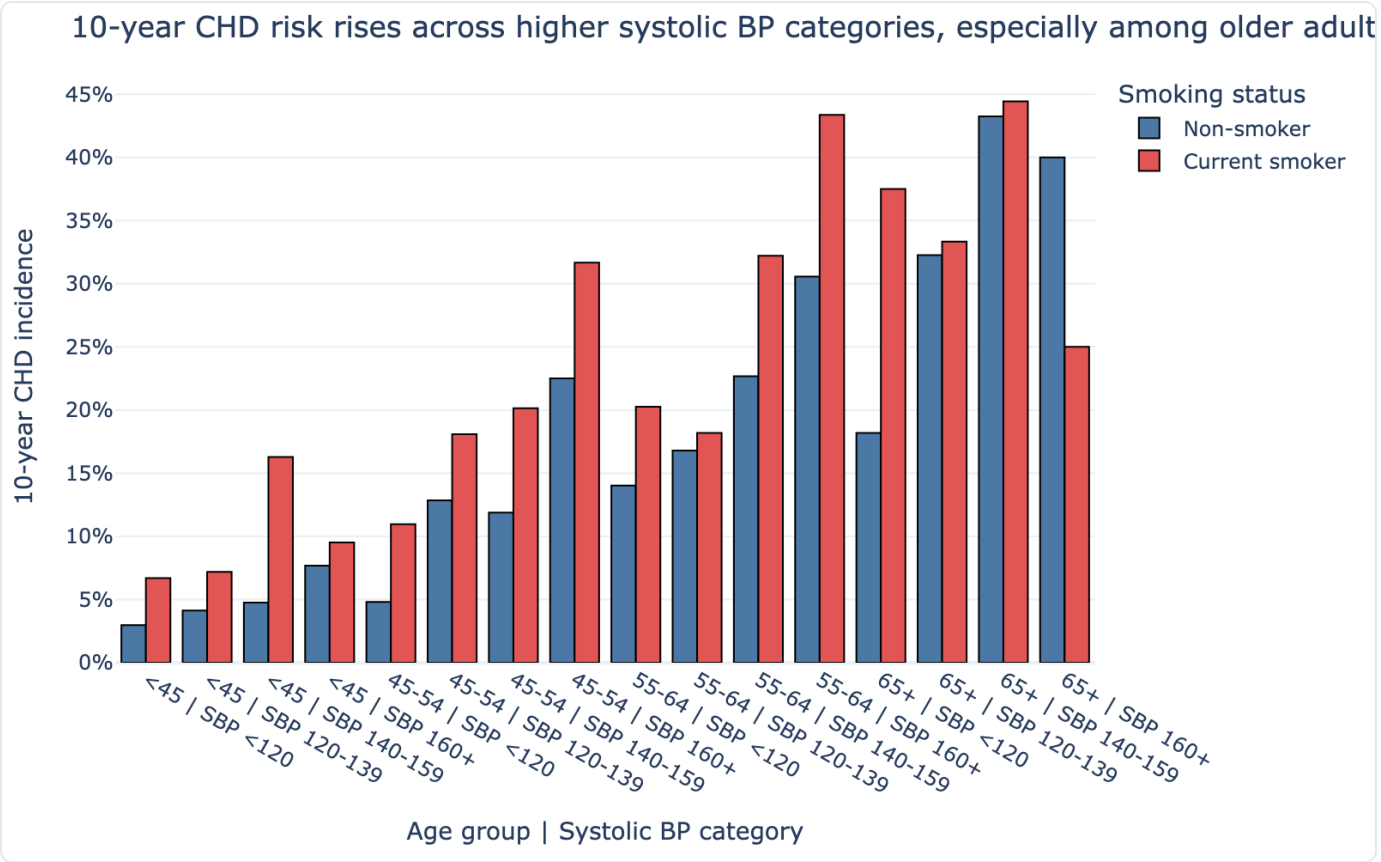

Figure 1

shows CHD risk climbing across higher systolic BP categories, with consistently higher risk among older adults and smokers. That suggests the later multivariable models should pay special attention to age, systolic BP, hypertension status, diabetes, and smoking specification.

Next Steps

Next I'll fit multivariable logistic regression models to identify which of these factors remain independently associated with 10-year CHD after adjustment, check collinearity among overlapping predictors like systolic BP vs diastolic BP and smoking status vs cigarettes per day, and then examine whether clinical threshold categories capture the risk pattern better than simple linear terms.

Independent CHD Risk Factors, Thresholds, and Subgroups

Overview

The strongest independent baseline risk factors for 10-year CHD in this cohort were age, male sex, smoking dose, systolic blood pressure, and glucose burden. The adjusted model summary `df_core_model_or` shows that after accounting for overlap among major predictors, CHD odds rose by about `1.93x` per 10 years of age, `1.67x` for men vs women, `1.24x` per 10 cigarettes/day, `1.14x` per 10 mmHg systolic BP, and `1.07x` per 10 mg/dL glucose.

Key Patterns & Observations

Age dominated the adjusted model. Every additional decade of age nearly doubled CHD odds, making it the clearest non-modifiable baseline risk factor. Male sex also remained substantial after adjustment, indicating that sex differences were not explained away by the measured risk profile.

Among modifiable factors, smoking dose and systolic BP stood out most clearly. Smoking intensity remained important even after adjustment, whereas a simple smoker/non-smoker distinction had looked weaker earlier. That means tobacco exposure appears better captured as dose rather than status alone. Systolic BP also remained independently important, even after accounting for prevalent hypertension and BP medication use, which suggests the measured BP level itself carries more information than the diagnosis label alone.

Metabolic risk was real, but partly shared across related variables. Glucose remained independently associated with CHD, but the diabetes indicator itself weakened substantially after adjustment, likely because glucose, BP, obesity, and hypertension history share overlapping information. Total cholesterol and BMI showed crude gradients but lost much of their independent strength in the full model, so in this cohort they appear to matter more as part of a broader cardiometabolic pattern than as stand-alone dominant predictors.

Risk clustering mattered a lot. When smoking, hypertension burden, diabetes, and obesity were counted together, CHD incidence rose from `8.3%` with zero of these factors to `14.6%` with one, `23.7%` with two, and `28.2%` with three. After adjustment for age and sex, the odds were still about `1.95x`, `2.80x`, and `4.03x` higher for 1, 2, and 3 factors versus none. The four-factor group had extremely high estimated risk, but it was based on only `5` people and should be treated cautiously.

Hypertension plus diabetes was a particularly high-risk cluster. CHD incidence was `10.5%` with neither condition, `23.3%` with hypertension alone, `26.5%` with diabetes alone, and `41.8%` with both. After adjustment for age, sex, and smoking, the odds ratio was `1.97` for hypertension alone, `2.07` for diabetes alone, and `4.42` for both together versus neither. Clinically, that means combined vascular and glycemic disease identifies a very high-risk subgroup even when each factor alone is already important.

Statistical Insights

The preferred multivariable model improved parsimony over the fuller specification, with lower `AIC` (`3004.79` vs `3008.56`) and `BIC` (`3092.86` vs `3109.22`). VIFs were all modest, with the highest around `3.71` for systolic BP and `2.97` for diastolic BP, so the main simplification was interpretive rather than a response to severe collinearity.

For the core model, the logistic structure was:

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 \text{Age} + \beta_2 \text{Male} + \beta_3 \text{Smoking dose} + \beta_4 \text{SBP} + \beta_5 \text{Glucose} + \cdots$$

This means each exponentiated coefficient is an adjusted odds ratio holding the other terms constant.

Threshold checks showed strong crude category gradients:

- Age-group incidence rose from `6.3%` (`<45`) to `36.6%` (`65+`)
- Systolic BP incidence rose from `8.3%` (`<120`) to `30.5%` (`>=160`)
- Glucose incidence rose from `14.1%` (`<100`) to `44.6%` (`>=126`)
- BMI incidence rose from `12.1%` (`18.5–24.9`) to `19.9%` (`>=30`)
- Cholesterol incidence rose from `10.4%` (`<200`) to `17.4%` (`>=240`)

But once adjusted, linear age and systolic BP terms performed about as well as or better than categorical or quadratic alternatives. Age quadratic slightly improved `AIC` by `0.60`, but not enough to make a compelling case for replacing the simpler linear term, and systolic BP quadratic was not favored. So the clinical takeaway is: **risk rises meaningfully across guideline thresholds, but the underlying adjusted relationship for age and systolic BP is still well approximated as progressive rather than sharply discontinuous.**

Subgroup testing found that most major risk factors were broadly similar in men and women. Smoking, hypertension burden, diabetes, and obesity did not show strong sex interactions. The notable exception was a high systolic BP category (`>=160 mmHg`), where the male interaction term was `1.81` (`95% CI 1.11–2.94` , `p=0.0166`), suggesting severe systolic hypertension may carry an even larger relative penalty in men.

Interpretation & Implications

For prevention, the most actionable modifiable targets are smoking exposure and systolic blood pressure, with glycemic burden close behind. Age and sex are the strongest contextual risk markers, but they are not modifiable. In contrast, smoking and BP both retained clear independent associations after adjustment, and both also contributed to strong risk clustering. That makes them especially valuable intervention targets because they are common, measurable, and clinically modifiable.

The predicted-risk chart **Figure 2**

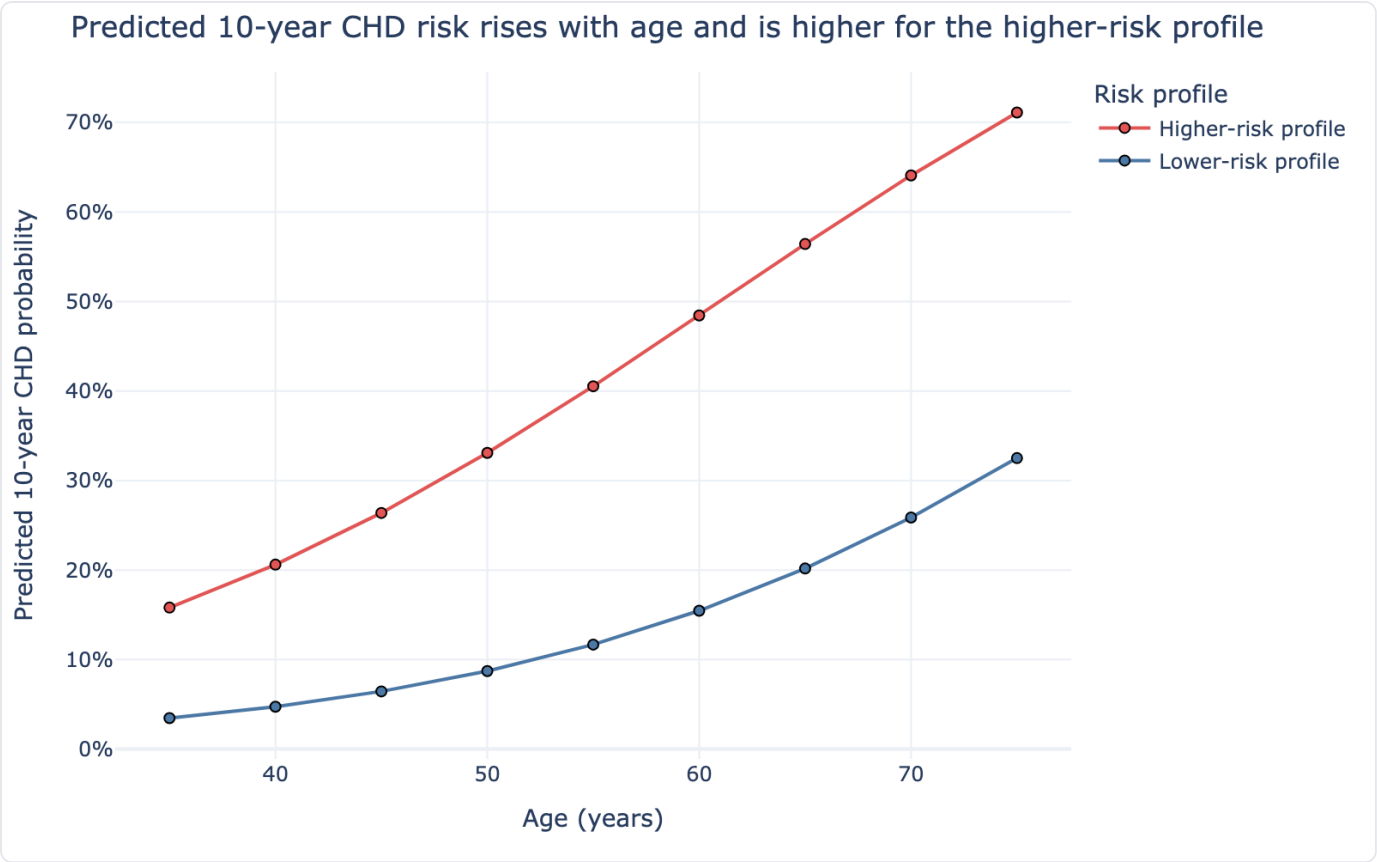

Figure 2

reinforces this visually: 10-year CHD probability rises with age, and the higher-risk profile stays above the lower-risk profile throughout adulthood. In practical terms, that means the same BP or smoking burden matters more as people get older because it compounds an already higher baseline risk.

The most important conceptual shift from the crude analysis is that some variables that look very strong alone partly reflect shared disease burden. Diabetes, prevalent hypertension, BMI, and cholesterol all matter clinically, but their independent coefficients shrink once age, systolic BP, smoking dose, glucose, and sex are considered simultaneously. So if the question is "what are the strongest independent risk factors," the answer is not simply "all traditional factors equally"; it is that **age sets the baseline risk, while smoking dose, systolic BP, and glucose remain the clearest independent modifiable signals.**

The next step is to convert these adjusted effects into a ranked clinical summary with example absolute risks and practical prevention scenarios, so the final answer can directly compare which factors matter most and how much risk might fall with realistic changes in smoking or blood pressure.

Strongest Baseline Risk Factors for 10-Year CHD

Overview

The strongest independent baseline risk factors in this Framingham cohort were older age, male sex, heavier smoking, higher systolic blood pressure, and higher glucose. The ranked adjusted-risk summary `df_chd_risk_summary` and the underlying final model results `df_core_model_or` show that

CHD odds rose by about `1.93x` per 10 years of age, `1.67x` for men vs women, `1.24x` per 10 cigarettes/day, `1.14x` per 10 mmHg systolic BP, and `1.07x` per 10 mg/dL glucose.

Key Patterns & Observations

Age was the dominant non-modifiable risk marker. Every additional decade nearly doubled the odds of 10-year CHD, which means baseline age is the main way overall risk gets set before any modifiable factor is considered.

Among modifiable factors, smoking burden and systolic BP were the clearest prevention targets. Smoking remained important after adjustment, especially when measured as cigarette dose rather than just smoker/non-smoker status. Systolic BP also stayed independently important even after accounting for prevalent hypertension and BP medication use, so the measured BP level itself mattered more than the diagnosis label alone.

Glucose added independent metabolic risk, but diabetes as a label became less informative once overlap was accounted for. That pattern suggests the model is capturing glycemic burden more directly through glucose than through the diabetes indicator by itself. Cholesterol and BMI still trended in the expected direction, but their adjusted effects were modest compared with age, smoking, systolic BP, and glucose.

Absolute risk differences were large for realistic patient profiles. A lower-risk illustrative profile had a predicted 10-year CHD probability of about `4.8%`, while a higher-risk profile reached about `63.0%`. Within that higher-risk profile, removing smoking lowered predicted risk to `52.5%`, lowering systolic BP reduced it to `49.9%`, and doing both lowered it further to `39.2%`. That translates to approximate absolute risk reductions of `10.5`, `13.1`, and `23.8` percentage points, corresponding to illustrative NNT values of about `10`, `8`, and `4` respectively if those changes were fully achieved and causal.

Statistical Insights The final adjusted model can be written as:

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 \text{Age} + \beta_2 \text{Male} + \beta_3 \text{Smoking dose} + \beta_4 \text{SBP} + \beta_5 \text{Glucose} + \cdots$$

The main adjusted effects were:

Risk factor	Comparison	Adjusted OR (95% CI)	Clinical meaning
Age	Per 10 years	1.93 (1.70, 2.18)	Strongest overall independent predictor
Male sex	Male vs female	1.67 (1.36, 2.04)	Men had substantially higher CHD odds
Smoking intensity	Per 10 cigarettes/day	1.24 (1.15, 1.34)	Clear modifiable dose-response signal
Systolic BP	Per 10 mmHg	1.14 (1.08, 1.21)	Important independent BP-related risk
Glucose	Per 10 mg/dL	1.07 (1.03, 1.12)	Independent metabolic risk signal
Prevalent hypertension	Yes vs no	1.23 (0.95, 1.59)	Elevated but less distinct after adjustment
Diabetes	Yes vs no	1.04 (0.57, 1.92)	Crude risk high, but adjusted estimate imprecise

The adjusted OR plot **Figure 3**

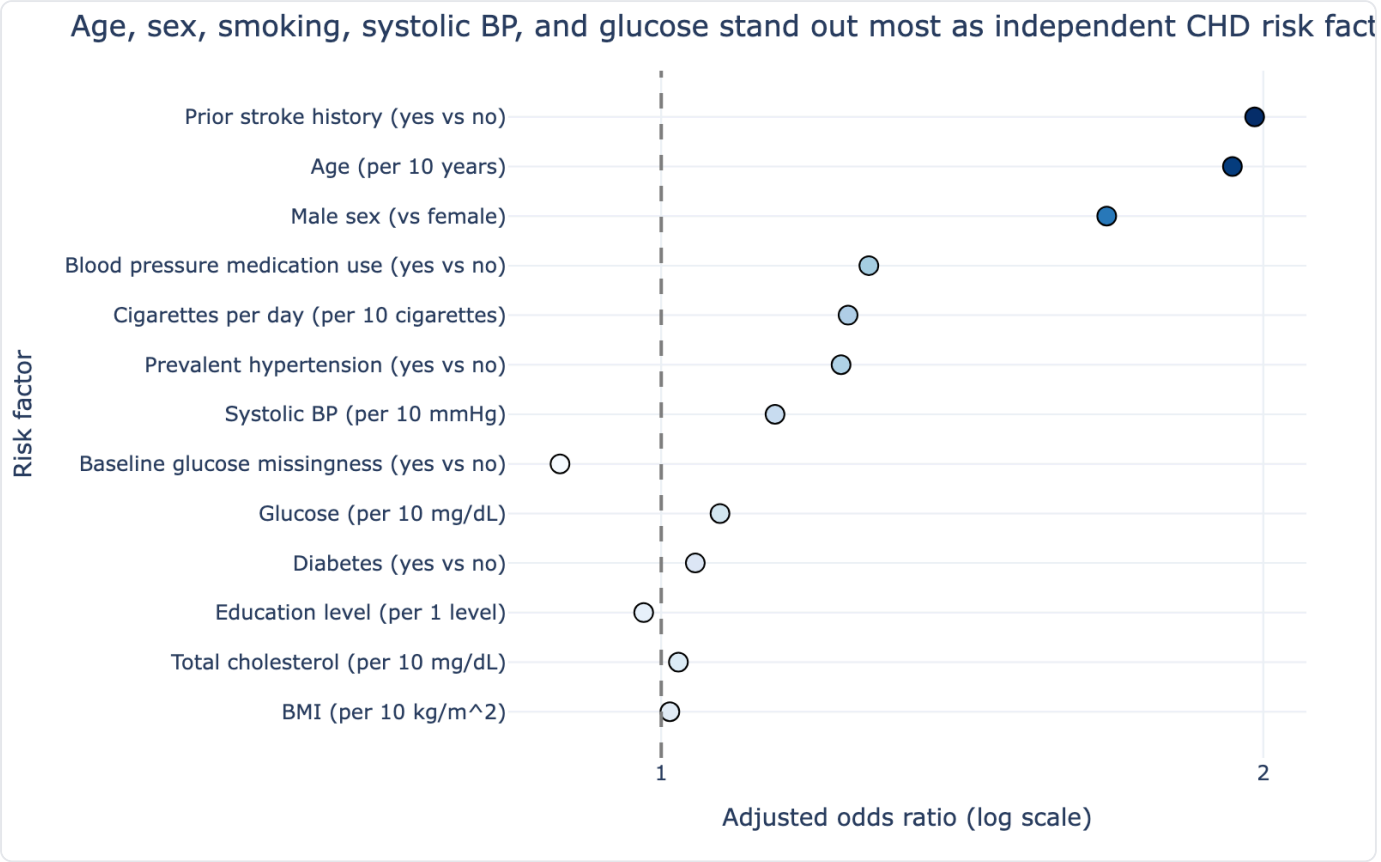

Figure 3

makes the ranking visually clear: age stands out first, followed by male sex, then smoking, systolic BP, and glucose as the main independent modifiable signals.

A targeted robustness check supported the main conclusions. Age, male sex, systolic BP, and glucose changed very little in a sensitivity model, while smoking remained positively associated but shifted in magnitude when coded as current smoker rather than cigarettes/day. That means the *presence* of a smoking effect is robust, while its exact size depends on how tobacco exposure is represented.

Interpretation & Implications

For prevention and risk stratification, the priority modifiable factors are smoking exposure and systolic blood pressure, with glycemic burden as a secondary but still important target. Age and sex explain who starts at higher baseline risk, but smoking and BP are the clearest levers for reducing that risk in practice.

This also helps reconcile the earlier crude and adjusted findings. Diabetes, hypertension history, BMI, and cholesterol all looked important in simpler comparisons, but much of their signal overlaps with age, measured systolic BP, smoking intensity, and glucose. So the final answer is not that traditional risk factors disappear; it is that **the strongest independent baseline predictors are age, male sex, smoking dose, systolic BP, and glucose, while the others mainly reinforce clustered cardiometabolic risk rather than dominate on their own.**

Limitations

This is an observational cohort analysis, so the model estimates association rather than proof of causation. Odds ratios may overstate relative risk somewhat because the outcome is not rare (~15%). Some variables were sparse, especially prior stroke and diabetes, which makes those estimates less precise. Glucose was partly imputed with a missingness flag, which preserved sample size well but still means glycemic findings should be interpreted as cohort-specific rather than definitive. The predicted intervention scenarios are illustrative model-based approximations, not trial-based treatment effects.

Bottom line: in this cohort, age sets the baseline, but the most actionable independent CHD risk factors are heavier smoking and higher systolic blood pressure, with higher glucose adding additional risk—so prevention decisions should focus first on tobacco exposure, BP control, and glycemic risk in people already entering older age or higher-risk sex strata.

Data Sources

- df_framingham_model
- df_core_model_or
- df_chd_risk_summary